AI651: Deep Learning for Space, Time and Graphs
Assignment 1 Report

Israr Ali, 25280102

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GitHub repository: https://github.com/IsrarAliKhan/AI651-PA-01

Conventions.

- Task 1 results come from the `PA1_PRESET=full` notebook run (data seed 0, model seed 0).
- Errors are validation RMSE in $\mathrm{m/s^2}$ unless stated otherwise.
- "Established" means Stations 1–3; "held-out" means Station 4.

Task 1: Forecasting Across Heterogeneous Sensors

Implementation. I completed `RawAttentionForecaster.forward`, `SeriesDecomposition.forward` (replicate padding) and `aggregate_delays` (index $(t - \tau_j) \bmod L$). All notebook checks pass.

Response 1A

Prior ordering

- Stations 1-3: Raw Attention < Period-routed < Sensor-specific < Shared
- Station 4: Raw Attention < Period-routed < Shared

Model	Established	A	B	C	Station 4	Params
Shared ridge	0.767	0.805	0.854	0.621	0.876	4,608
Sensor-specific ridge	0.768	0.804	0.852	0.630	n/a	13,824
Period-routed ridge	**0.166**	0.181	0.164	0.152	**0.173**	59,904
Raw Attention	0.421	0.380	0.475	0.403	0.608	368,368

Table 1: Output 1.1, validation RMSE ($\mathrm{m/s^2}$). A/B/C are established stations by product.

Measured.

- Period routing cuts shared-ridge error by 78% (0.767 → 0.166).
- Per-station matrices give no gain (0.768) and cannot serve Station 4.
- Raw Attention beats the plain ridges but is 2.5× worse than period routing, with 6× more parameters.
- Held-out degradation: period routing +4%, Raw Attention +44%.

Revised ordering: Period-routed < Raw Attention < Shared ≈ Sensor-specific.

Why.

- Shared and sensor-specific ridge assume one map per window or per station. Every station sees every pace (14–36 samples), so both must average incompatible continuations.
- Period routing conditions on pace, the true generative variable, so it also transfers to Station 4.
- Raw Attention must learn pace while also modelling drift. It fits station-specific features, hence its large held-out gap.

Response 1B

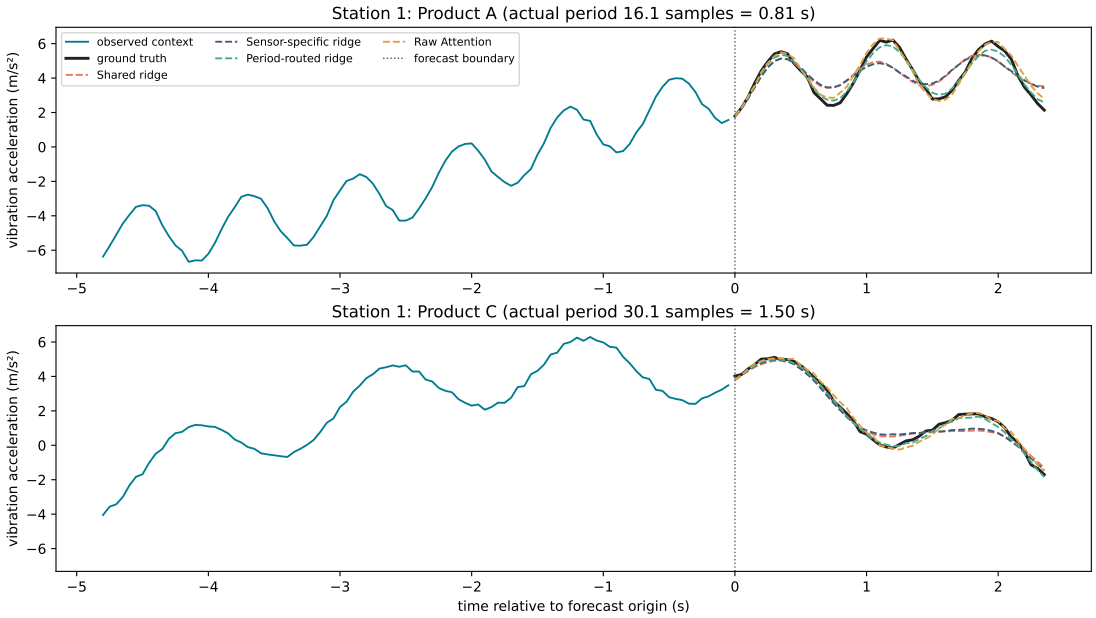


Figure 1: Output 1.2: Station 1 forecasts for Product A ($P = 16.1$) and Product C ($P = 30.1$).

The compromise. $\widehat{W}$ is one least-squares map over all paces. It has no input that selects the right phase advance, so it predicts the pace-averaged continuation. Averaging out-of-phase cycles shrinks amplitude:

- Product A: the forecast peaks at about 5.0 vs a true 6.2 (window RMSE 0.779 vs 0.232 for period routing).
- Product C: the second hump is halved, about 0.9 vs 1.8 (0.448 vs 0.150).

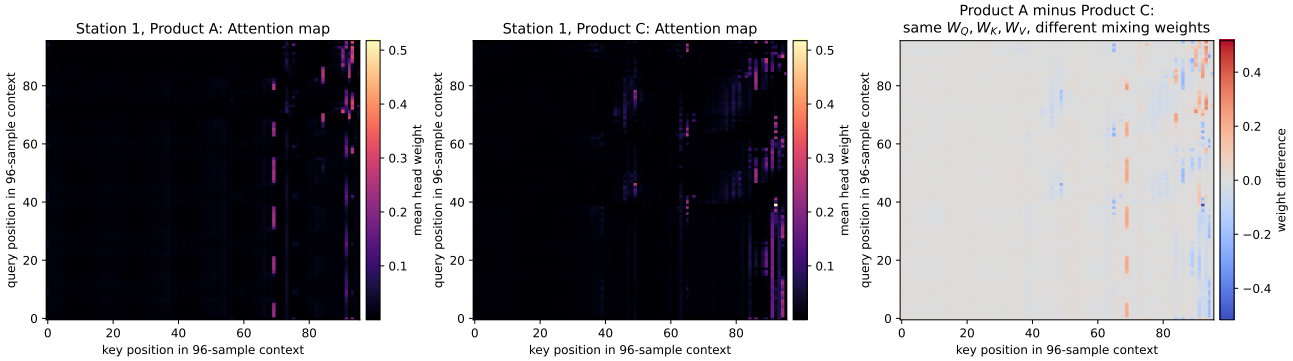


Figure 2: Output 1.3: attention maps for the Product A and C contexts and their difference.

Context-built mixing. With the same frozen W_Q, W_K, W_V:

- the A map focuses on key 69 and the last keys, while the C map spreads over keys 40–95;
- weights differ by up to 0.52.

Because $Q = XW_Q$ and $K = XW_K$, a new context changes $\text{softmax}(QK^\top/\sqrt{d_h})$ and therefore the mixing operator. Ridge applies one fixed $\widehat{W}$ to every input. This shows context dependence, not period recovery.

Response 2

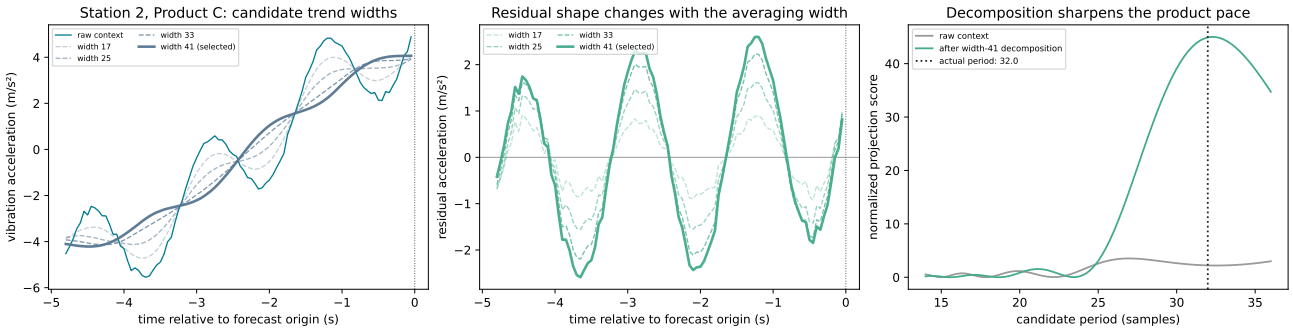


Figure 3: Output 2.1: trend and residual for candidate widths (Station 2, Product C, $P = 32$).

Selected width: $m = 41$.

- **Visual (2.1).** Small widths leave vibration in the trend. At $m = 41$ the trend is a smooth ramp and the residual is a clean cycle (std $0.58 \to 1.57\,\text{m/s}^2$). The period score peaks sharply at the true $P = 32$ only after decomposition.
- **Oracle (2.2).** Period recovery within one sample rises from 0.687 (raw) to 0.980 ($m = 17$) and 0.998 ($m = 41$).
- **Ablation (2.3).** Raw Attention $0.421 \to 0.287$ established (-32%) and $0.608 \to 0.428$ held-out (-30%). The gain is largest on steep-ramp contexts.

Why a moving average fits. Drift ($\sim$240 samples) and vibration (14–36) have separated time scales. A centered average longer than P removes the cycle and keeps the near-linear drift. The parameter-free filter cannot overfit station drift, which matches the held-out gain.

Why the oracle diagnostic. It measures directly what m should achieve (exposing the period). It uses the true period only to score widths on training windows, so there is no leakage.

Failure case. An abrupt level shift is smeared into a ramp and leaves a false bump in S. A median filter or STL would preserve jumps and curvature, at the cost of more hyperparameters.

Response 3

Verification.

- The worked example gives $z = [35, 15, 25, 25]$, as expected, confirming that delays look back.
- `delay_scores` matches direct correlation $(4, -4, 4, -4)$.

Strongest lags.

- Residual correlation peaks at $\tau = 16$ (0.896) and $\tau = 32$ (0.897), i.e. P and $2P$.
- Troughs fall at half-period offsets (24, 40).

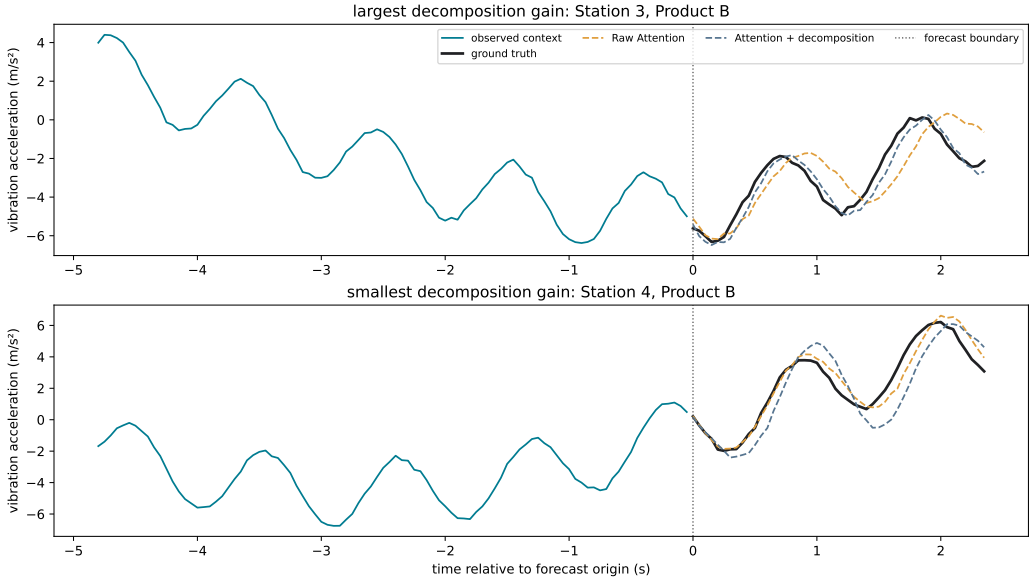


Figure 4: Output 2.3: largest gain on a steep ramp (top), smallest on a flat context (bottom).

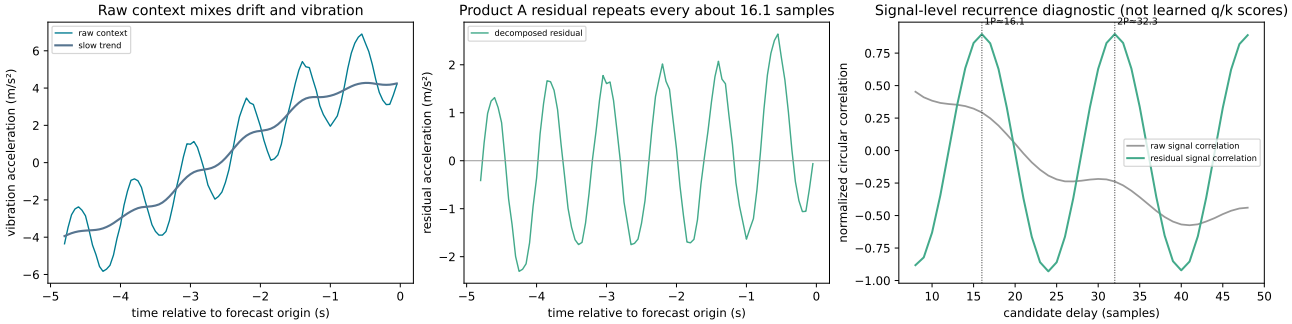


Figure 5: Output 3.2: raw vs residual recurrence (Station 2, Product A, $P = 16.1$).

- A periodic signal recurs at every multiple of P, so blending these lags is safe.

Why decomposition helps. The raw correlation follows the ramp: it is high at short lags and falls steadily, with no peak at P. Removing drift leaves a zero-mean cycle that correlates only when peaks align. Useful structure is then a few shared displacements, which is exactly what the delay mixer models with L scores instead of L^2.

Failure case. If the period drifts within the context (e.g. 14 to 20 samples), no single lag aligns the whole window. Blending mismatched lags cancels the cycle and damps the forecast. Isolated spikes also fail, because circular shifts create false echoes.

Response 4

(a) Prediction: decomposition should help most when drift is prominent, because it hands the ramp to a linear head before mixing.

(b) Results.

- **Each switch helps alone.** MSE gain vs Raw Attention: delay mixing 0.128, decomposition 0.096.
- **Combined gain is 0.139, far below the 0.224 sum.** The switches overlap because both target

Model	RMSE	MSE	St. 4 RMSE	St. 4 MSE	Params	Fit (s)
Shared ridge	0.767	0.588	0.876	0.767	4,608	0.002
Period-routed ridge	**0.166**	**0.028**	**0.173**	**0.030**	59,904	0.002
Raw Attention	0.421	0.178	0.608	0.369	368,368	802
Raw delay mixer	0.224	0.050	0.337	0.113	368,370	1097
Attention + decomposition	0.287	0.082	0.428	0.183	373,024	967
Autoformer-inspired	0.198	0.039	0.240	0.057	373,026	1017

Table 2: Outputs 1.1 and 4.1: validation error and cost (established unless St. 4).

drift plus repetition. The same holds on Station 4 (0.312 vs 0.442).

- **Prediction (a) was partly right.** Decomposition helped most on ramps, but delay mixing helped more overall.
- **Horizon (4.2).** Errors grow as phase errors accumulate. From $h = 1$ to $h = 48$: Raw Attention 0.149 → 0.530, Autoformer-inspired 0.135 → 0.235, period routing 0.112 → 0.228 (lowest throughout).

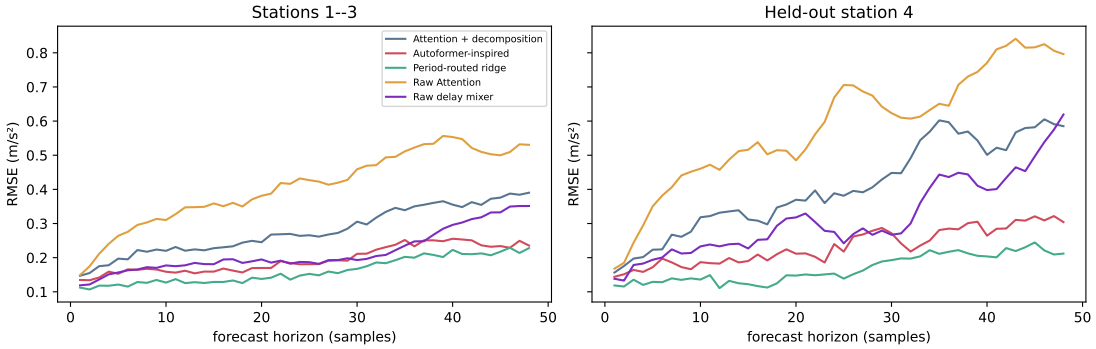


Figure 6: Output 4.2: RMSE along the 48-step horizon.

(c) Deployment. I chose **Period-routed ridge** for both populations, from validation only:

- Established: RMSE 0.166 vs 0.198 for the best neural model, 6× fewer parameters, 0.002 s vs 1017 s to fit.
- Station 4: RMSE 0.173 vs 0.240. It conditions on pace rather than station identity.
- Caveat: an unseen pace outside the routing bands would break it; Autoformer-inspired is the fallback.
- Test (Output 4.3): 0.161 established, 0.183 held-out.

Interpretation.

- **The right inductive bias mattered more than flexibility.** A linear map routed by pace beat every Attention variant with 6× fewer parameters, because it encodes the true cause directly.
- **The neural upgrades show the same lesson.** Decomposition and delay mixing each removed 30–50% of Raw Attention's error at no parameter cost, but overlapped when combined.
- **Flexibility without bias generalised worst.** Raw Attention lost 44% on Station 4.
- **Learned models pay off only where hand-built bias fails**, such as unseen paces.

Task 2: Leaderboard Challenge

Model	Autoformer, 1 encoder + 1 decoder layer, $d = 32$, future covariates
Parameters P / epochs E	22,081 / 4
Validation RMSE (3 seeds)	72.0 ± 2.9 (best baseline 112.0)
Leaderboard RMSE / score	**98.35 / 98.351 (rank 30, 1st attempt)**
Leaderboard MAE / sMAPE	66.30 / 74.61%

Code: `Question 2 - Leaderboard/autoformer.py` (modes `ablate`, `final`, `figures`).

Data and preprocessing

- **Target.** Non-negative and heavy-tailed: mean 98, std 91, max 994. Autocorrelation decays fast (0.97 at lag 1, 0.40 at lag 24, 0.03 at lag 168).
- **Period.** The periodogram has sharp peaks at 24 and 12. I treat one step as one hour and add sin / cos of (idx mod 24) as phase features (Fig. 7).
- **Covariates.** Six continuous variables (A–F) and a 4-way one-hot (G–J). A–C are z-scored; D–F are log-transformed then z-scored. Statistics use the training region only.
- **Scaling.** Global z-score of the target. Window-relative normalisation was avoided because absolute level carries signal (high levels tend to fall).

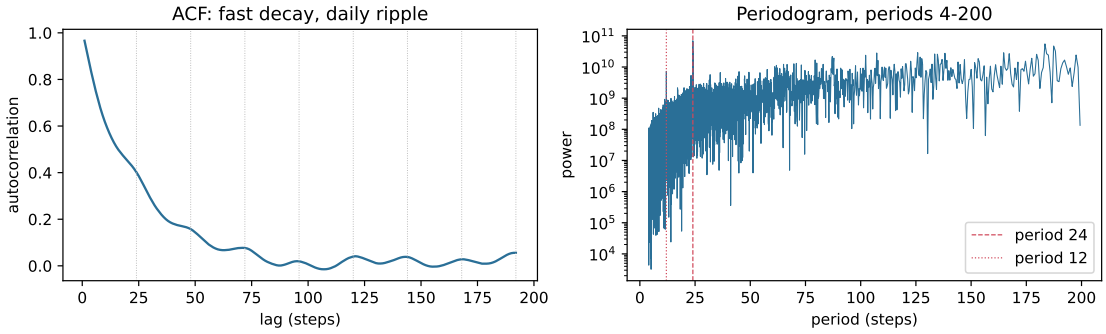


Figure 7: Autocorrelation and periodogram of the training series: a 24-step cycle.

Model

Written in PyTorch, adapted from the official implementation:
https://github.com/thuml/Autoformer. It contains both required mechanisms:

- **Series decomposition**: a moving average with kernel 25, applied at the input and after every sublayer. The decoder accumulates trend.
- **Auto-Correlation**: FFT delay scores, top $k = \lfloor \ln L \rfloor = 5$ lags, and time-delay aggregation. It replaces dot-product attention in self- and cross-attention.

Covariates and phase enter as "marks" in both the encoder (past) and the decoder (horizon). Input 168, label 48, output 168.

Validation protocol

- **Chronological split.** The last 4,368 points form validation: 176 origins, each forecasting a full 168-step block. Training targets end before this region.

- **Training.** MSE loss, Adam (10^{-3}, halved each epoch), early stopping (patience 2). Each epoch sees every 4th window.
- **Seeds and metrics.** Three seeds per configuration; RMSE, MAE and sMAPE reported.

Ablation

Configuration	RMSE (mean ± sd)	MAE	sMAPE (%)
Last-24h mean (best baseline)	112.0	80.6	88.8
No covariates	95.3 ± 0.3	72.4	81.1
No covariates, log target	96.5 ± 0.1	67.6	82.9
Covariates, past only	92.4 ± 1.0	70.2	80.1
Covariates incl. horizon	$\mathbf{72.0 \pm 2.9}$	54.6	71.7
Covariates incl. horizon, log target	75.0 ± 3.6	**48.8**	**59.4**

Table 3: Validation results, seeds 0–2.

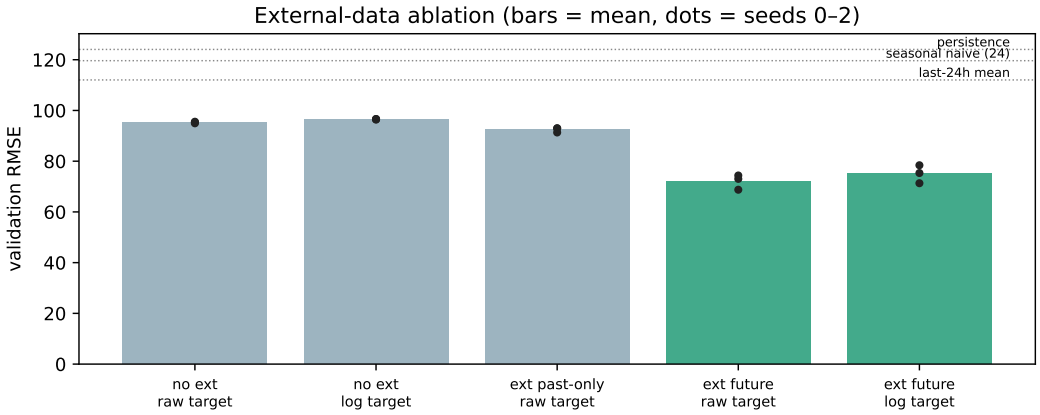


Figure 8: Ablation over seeds; dotted lines are baselines.

- **Future covariates matter, past ones barely do.** Past-only covariates reduce RMSE by 2.9; covariates over the horizon reduce it by 24%, about 8× the seed spread. Weather during the horizon decides whether the level builds or clears (Fig. 9).
- **Log target trades RMSE for MAE.** It improves MAE and sMAPE but underpredicts spikes, which dominate RMSE. Ranking uses RMSE, so I kept the raw target.
- **Final model.** Retrained from scratch on the full history with the best configuration, seed 0, $E = 4$ (the mean best epoch).

Use of Generative AI

I used Claude Code (Anthropic, Claude Opus 5.5). I asked it to explain the Autoformer architecture and the series decomposition mechanism, and to summarize the differences between attention and autocorrelation. I used it to understand the tasks and how should I approach them. I used it for scaffolding the tasks code and drafting the LaTeX report.

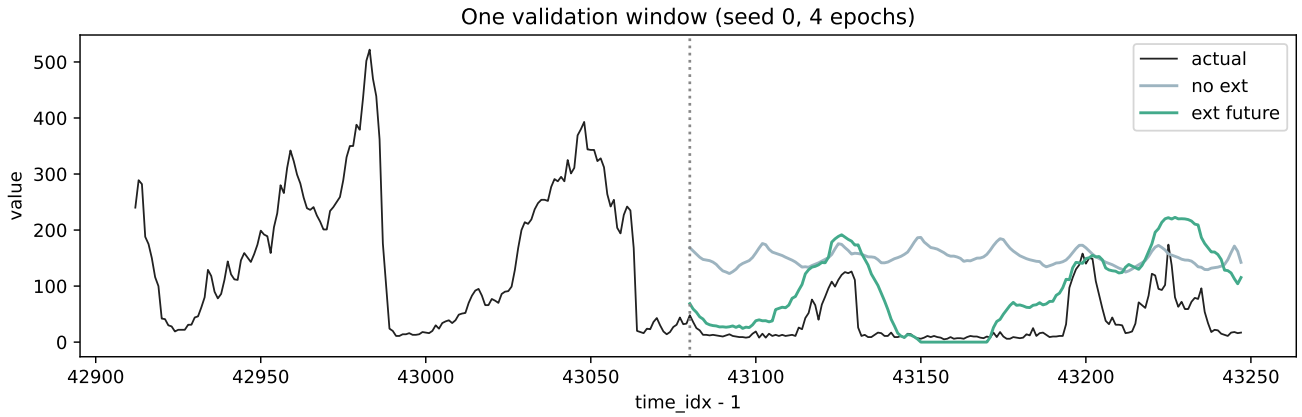


Figure 9: One validation window. Horizon covariates let the model anticipate drops and rises.

References

- Wu et al. (2021). Autoformer: Decomposition Transformers with Auto-Correlation for Long-Term Series Forecasting. NeurIPS. Code: https://github.com/thuml/Autoformer.
- Vaswani et al. (2017). Attention Is All You Need. NeurIPS.
- Zeng et al. (2023). Are Transformers Effective for Time Series Forecasting? AAAI.